Engineering Rubric — Tabletop Alert / Signal Device

Chapter 6 · Grade 4 · 15 points (aligned to AP 13.4)

Name: _____ Date: _____

DESIGN — 5 pts

What to look for	Full credit	Partial credit
Blueprint	clear, labeled (trigger, output, energy change)	some labels missing
Criteria	all criteria listed and addressed	some criteria missing
Constraints	all constraints listed and respected	constraints partly respected
Materials	materials listed and used as planned	materials listed but not matched to design
Problem	problem restated clearly	problem unclear

ENERGY MODEL — 3 pts

What to look for	Full credit	Partial credit
Input/output	correct initial AND final forms of energy	one form incorrect
Transformation	correctly identifies the energy transformation	transformation missing
Observable change	explains the observable output/change	output unclear

TESTING / REDESIGN — 3 pts

What to look for	Full credit	Partial credit
Test 1	records Test 1 results (3 trials)	Test 1 partly recorded
Evidence-based change	changes ONE thing with a reason	change without a reason
Test 2	records Test 2 and compares to Test 1	Test 2 incomplete

FINAL EXPLANATION — 4 pts

What to look for	Full credit	Partial credit
Claim/conclusion	clear claim about the device	claim unclear
Evidence	uses observed evidence (not “proves”)	little or no evidence
Reasoning	links evidence to the claim	reasoning weak
Engineering vocabulary	uses criteria/constraints/test/refine/energy correctly	vocabulary missing

Total: ____ / 15

Note: a failed prototype is NOT a penalty — score the reasoning and evidence, not the decoration.